Week 2:
Basic Probability

OUTLINE

- Review of set theory
- Assigning and determining probabilities
- Properties of the probability function
- Probabilities for continuous sample spaces
- Probabilities for finite sample spaces
- Combinatorics
- Binomial probability law
- Real-world example: quality control

Review of Set Theory

- A *set* is a collection of objects.
- A set can be defined by either the *enumeration method* or the *description method*.
- Example: Let A be the set of natural numbers. Specify A with the enumeration and description methods

$$A = \{1,2,3, \dots\} \quad \text{(enumeration)}$$
$$A = \{x: x \text{ is an integer and } x \geq 1\} \quad \text{(description)}$$

- Each object in the set is called an *element*. An element is listed only once and ordering of elements within the set is not important.
- An element x of a set A is denoted by the notation $x \in A$ and is read as x is contained in A.

Review of Set Theory

- A set having no elements is called the *empty* or *null* set and is denoted by $\emptyset$.
- The set of all objects is called the *universal* set and is denoted by S.
- A *simple* set is a set containing a single element.
- A set B is defined as a *subset* of a set A if every element in B is also an element of A. The notation for subset is $B \subset A$.
- Two sets are *equal* if they contain the same elements. This is denoted by $A = B$. Note that if $B \subset A$ and $A \subset B$, then $A = B$ (why?)
- The *complement* of A, denoted by A^c, is the set of elements in the universal set, but not in A

$$A^c = \{x : x \notin A,\ x \in S\}.$$

Review of Set Theory

- The *union* of *A* and *B*, denoted by $A \cup B$, is the set of elements that belong to *A* or *B* or both *A* and *B*

$$A \cup B = \{x: x \in A \text{ or } x \in B\}$$

- The *intersection* of *A* and *B*, denoted by $A \cap B$, is the set of elements that belong to both *A* and *B*

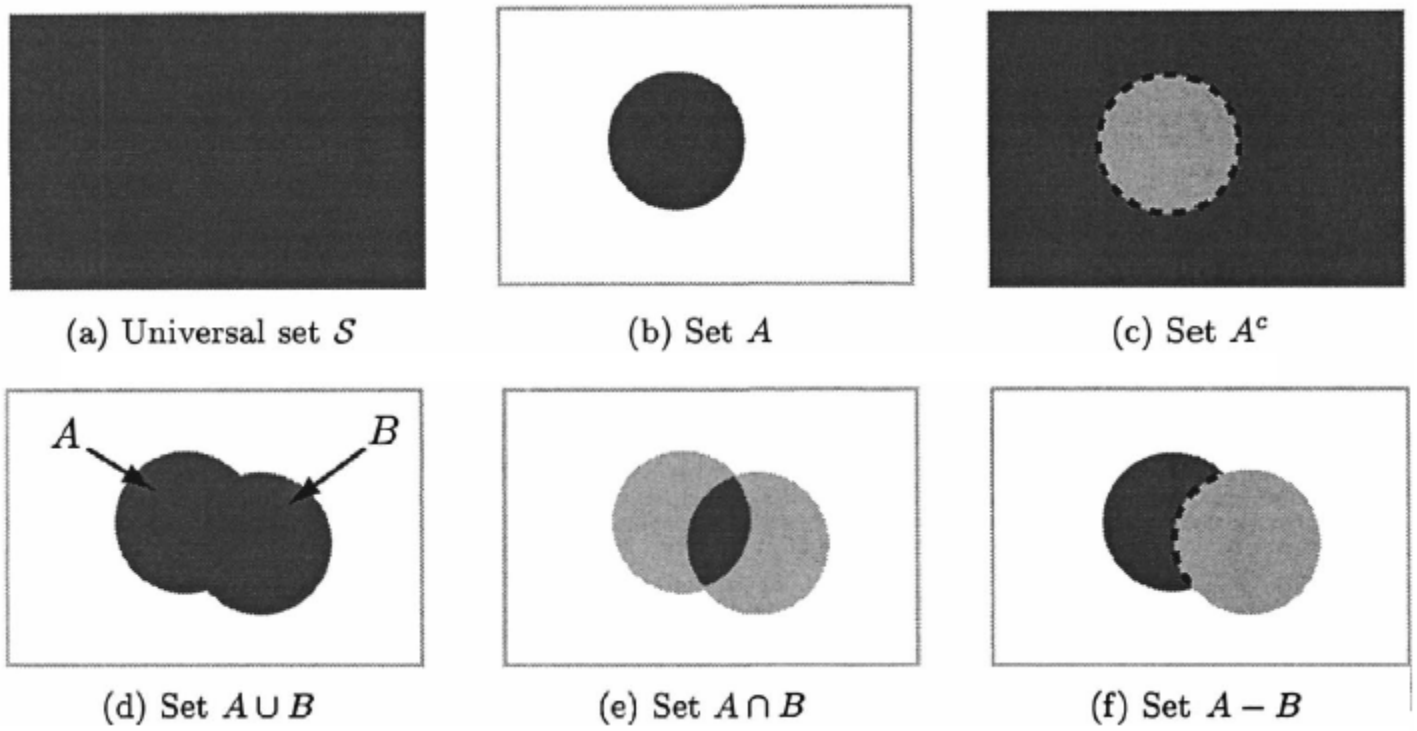
$$A \cap B = \{x: x \in A \text{ and } x \in B\}$$

- The *difference* between sets, denoted by $A - B$, is the set of elements in *A*, but not in *B*

$$A - B = \{x: x \in A,\ x \notin B\}$$

Review of Set Theory

Operations on sets can be illustrated pictorially using a *Venn diagram*.



(a) Universal set S (b) Set A (c) Set A^c

(d) Set $A \cup B$ (e) Set $A \cap B$ (f) Set $A - B$

Figure source: S. M. Kay, *Intuitive Probabilty and Random Processes Using MATLAB*, Prentice Hall, 2006.

Review of Set Theory

- The following results hold for the sets:

$$(A^c)^c = A$$
$$A^c \cup A = S, A^c \cap A = \emptyset$$
$$A \cup \emptyset = A,\ A \cap \emptyset = \emptyset$$
$$A \cup S = S, A \cap S = A$$
$$S^c = \emptyset,\ \emptyset^c = S$$

- If two sets A and B have no elements in common, they are said to be *disjoint*. What is the condition for being disjoint?

- Sets $A_1, \ldots, A_N$ are said to be *mutually disjoint* if $A_i \cap A_j = \emptyset$ for all $i \neq j$.

- Mutually disjoint sets $A_1, \ldots, A_N$ are said to *partition* the universal set if

$$S = A_1 \cup A_2 \cup \cdots \cup A_N$$

Review of Set Theory

Algebraic rules for manipulating multiple sets:

Commutativity:

$$A \cup B = B \cup A, \qquad A \cap B = B \cap A.$$

Associativity :

$$(A \cup B) \cup C = A \cup (B \cup C),$$
$$(A \cap B) \cap C = A \cap (B \cap C)$$

Distributivity :

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

De Morgan's Law:

$$A \cup B = (A^c \cap B^c)^c$$
$$A \cap B = (A^c \cup B^c)^c$$

Review of Set Theory

- A set is said to be *discrete* if the number of its elements can be counted.
- Discrete sets can have finite or infinite number of elements. In the former case, the set is called a *finite set*. In the latter case, the set set is said to be a *countably infinite set*.
- A set is said to be *continuous* if the number of its elements can not be counted.
- Example:

$$A = \left\{\frac{1}{8}, \frac{1}{4}, \frac{1}{2}, 1\right\}, \text{finite set} - \text{discrete}$$

$$B = \left\{1, \frac{1}{2}, \frac{1}{3}, \ldots\right\}, \text{countably infinite set} - \text{discrete}$$

$$C = \{x: 0 \leq x \leq 1\}, \text{infinite set} - \text{continuous}$$

Assigning and Determining Probabilities

- The universal set is termed the *sample space* and its elements are the *outcomes* or the *sample points*.

- A subset of the sample space is termed an *event*. Special events are

  simple events: events containing only a single outcome
  certian event: the sample space
  impossible event: the null set

- Two events are said to be *mutually exclusive* if both events can not occur simultaneously.

Conclusion: Probabilistic questions can be formulated using the set theory with its own terminology. Equivalent terms are given below in a table.

Assigning and Determining Probabilities

Set theory	Probability theory	Probability symbol
universe	sample space (certain event)	$\mathcal{S}$
element	outcome (sample point)	s
subset	event	E
disjoint sets	mutually exclusive events	$E_1 \cap E_2 = \emptyset$
null set	impossible event	$\emptyset$
simple set	simple event	$E = \{s\}$

Table source: S. M. Kay, *Intuitive Probabilty and Random Processes Using MATLAB*, Prentice Hall, 2006.

Assigning and Determining Probabilities

- In order to develop a theory of probability, we must assign probabilities to events.
- *Probability function* assigns a number between 0 and 1 to events.
- In the *axiomatic approach to probability*, the following axioms are assumed to hold for the probability function

Axiom 1: $P[E] \geq 0$ for any event E
Axiom 2: $P[S] = 1$
Axiom 3: If A and B are mutually exclusive events, $P[A \cup B] = P[A] + P[B]$
Axiom 3′: $P\left[\bigcup_{i=1}^{N} E_i\right] = \sum_{i=1}^{N} P[E_i]$
Axiom 4: $P[\bigcup_{i=1}^{\infty} E_i] = \sum_{i=1}^{\infty} P[E_i]$

Assigning and Determining Probabilities

Example: Determine the probability that the outcome of a fair die toss is even. Assume that the outcomes are equally likely

The simple events : $\{1\}, \{2\}, \{3\}, \{4\}, \{5\}, \{6\}$
The given event E : $\{2,4,6\}$
Let E_i represent the simple event $\{i\}$. Then, $E = E_2 \cup E_4 \cup E_6$.

From Axiom 2: $P[S] = P\left[\bigcup_{i=1}^{6} E_i\right] = 1$

From Axiom 3': $P\left[\bigcup_{i=1}^{6} E_i\right] = \sum_{i=1}^{6} E_i = 1$ (simple events are mutually exclusive!)

Let $P[E_1] = P[E_2] = \cdots P[E_6] = p$. Then, $\sum_{i=1}^{6} E_i = \sum_{i=1}^{6} p = 6p = 1 \Rightarrow p = 1/6$.

From Axiom 3': $P[E] = P[E_2 \cup E_4 \cup E_6] = P[E_2] + P[E_4] + P[E_6] = 3/6 = 1/2$

Assigning and Determining Probabilities

Example: A person arrives at the theater late by s_i minutes, where

$$s_i = 1,2,\cdots, \qquad P[s_i] = (1/2)^i$$

What is the probability that the person will be late more than 1 minute?

The simple events : $\{1\}, \{2\}, \cdots$
The given event E : $\{2,3,4,\cdots\}$
Let E_i represent the simple event $\{i\}$. Then, $E = \bigcup_{i=2}^{\infty} E_i$

From Axiom 4: $P[E] = P[\bigcup_{i=2}^{\infty} E_i] = \sum_{i=2}^{\infty} P[E_i] = \sum_{i=2}^{\infty} (1/2)^i$

$$P[E] = \frac{\left(\frac{1}{2}\right)^2}{1-\frac{1}{2}} = \frac{1}{2}, \qquad \left(\sum_{i=k}^{\infty} a^i = \frac{a^k}{1-a}\right)$$

Properties of the Probability Function

Property 1: $P[E^c] = 1 - P[E]$.

Proof: By definition, $S = E \cup E^c$. Also, E and E^c are mutually exclusive.

$$\begin{aligned} 1 &= P[S] \\ &= P[E \cup E^c] \\ &= P[E] + P[E^c] \\ &\Rightarrow P[E^c] = 1 - P[E] \end{aligned}$$

Property 2: $P[\emptyset] = 0$.

Proof: $\emptyset = S^c$. Hence, we have

$$\begin{aligned} P[\emptyset] &= P[S^c] \\ &= 1 - P[S] \\ &= 1 - 1 \\ &= 0 \end{aligned}$$

Properties of the Probability Function

Property 3: $0 \leq P[E] \leq 1$ for any E.

Proof:

For any E, from Axiom 1, $P[E] \geq 0$. Also,

$$\begin{aligned} S &= E \cup E^c \\ P[S] &= P[E] + P[E^c] \\ 1 &= P[E] + P[E^c] \end{aligned}$$

From Axiom 1, $P[E^c] \geq 0$.

Hence,

$$P[E] = 1 - P[E^c] \leq 1.$$

Properties of the Probability Function

Property 4: $P[E \cup F] = P[E] + P[F] - P[E \cap F]$

Proof:

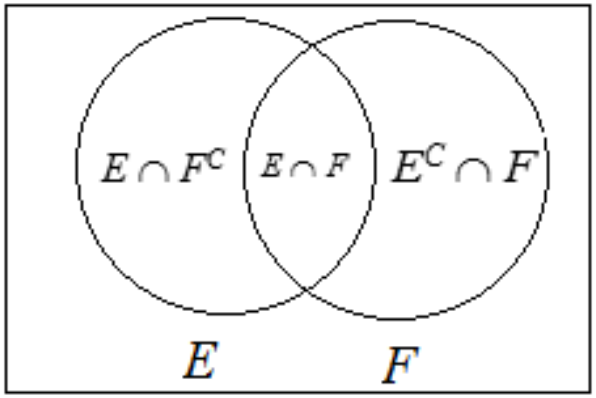
$$P[E \cup F] = P[E \cap F^C] + P[E \cap F] + P[E^C \cap F]$$

$$P[E] = P[E \cap F^C] + P[E \cap F] \Rightarrow P[E \cap F^C] = P[E] - P[E \cap F]$$

$$P[F] = P[F \cap E^C] + P[E \cap F] \Rightarrow P[F \cap E^C] = P[F] - P[E \cap F]$$

$$P[E \cup F] = P[E] - P[E \cap F] + P[E \cap F] + P[F] - P[E \cap F]$$

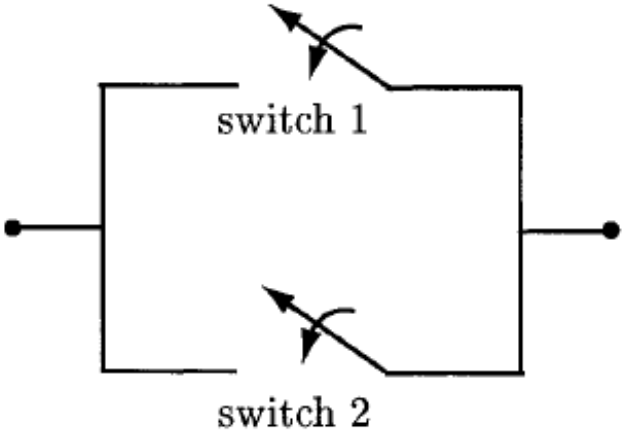
$$= P[E] + P[F] - P[E \cap F]$$



Properties of the Probability Function

Example: The circuit works if at least one of the switches is closed. Each switch has probability of 1/2 of being closed. The probability that both switches are closed is 1/4. What is the probability that the circuit will work?

E_1 ={switch 1 is closed}
E_2 ={switch 2 is closed}
E ={at least one switch is closed}= $E_1 \cup E_2$



$$P[E_1 \cup E_2] = P[E_1] + P[E_2] - P[E_1 \cap E_2]$$

$$= \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4}$$

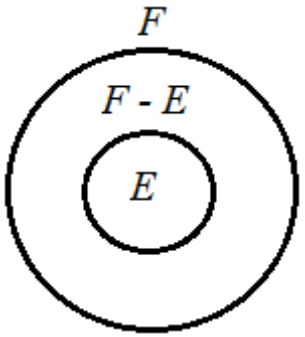
Figure source: S. M. Kay, *Intuitive Probabilty and Random Processes Using MATLAB*, Prentice Hall, 2006.

Properties of the Probability Function

Property 5: If $E \subset F$, then $P[E] \leq P[F]$

Proof:
If $E \subset F$, then $F = E \cup (F - E)$.
Since E and F - E are mutually exclusive

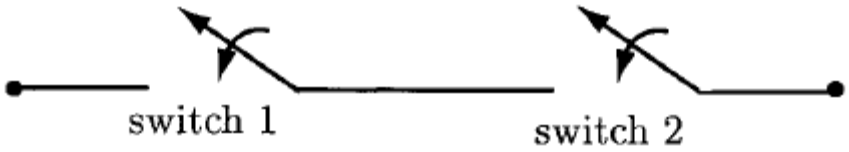
$$P[F] = P[E] + P[F - E]$$
$$\geq P[E]$$



One result: the probability of the intersection is less than or equal to the probability of the set with the smallest probability because $E \cap F \subset E$ and $E \cap F \subset F$ implies that $P[E \cap F] \leq P[E]$ and $P[E \cap F] \leq P[F]$.

Properties of the Probability Function

Example: The circuit works if both switches are closed. For the same switches as descriped in the parallel case, what is the probability that the circuit will work?



E_1 ={switch 1 is closed}
E_2 ={switch 2 is closed}
E ={both switches are closed}= $E_1 \cap E_2$

$$P[E_1 \cap E_2] = \frac{1}{4}$$

In which case (parallel or serial), the probability is higher? Why?

Figure source: S. M. Kay, *Intuitive Probabilty and Random Processes Using MATLAB*, Prentice Hall, 2006.

Properties of the Probability Function

The formula for the probability of the union of two events can be extended to more than two events. For example,

$$P[D \cup E \cup F] = P[D]+P[E]+P[F]-P[D\cap E]-P[D\cap F]-P[E\cap F]+P[D \cap E \cap F]$$

Proof:

$$\begin{aligned} P[D \cup E \cup F] &= P[D \cup (E \cup F)] \\ &= P[D] + P[E \cup F] - P[D \cap (E \cup F)] \\ &= P[D] + P[E] + P[F] - P[E \cap F] - P[(D \cap E) \cup (D \cap F)] \end{aligned}$$

Since $P[(D \cap E) \cup (D \cap F)] = P[D \cap E] + P[D \cap F] - P[D \cap E \cap F]$

$$P[D \cup E \cup F] = P[D]+P[E]+P[F]-P[D\cap E]-P[D\cap F]-P[E \cap F]+P[D \cap E \cap F]$$

Counting Methods

- We are interested in the number of choosing *r* objects out of *N* objects.
- Two possible strategies:

  Sampling with replacement: The object is placed back after having been choosen
  Sampling without replacement: It is obvious

- Also, the order in which the outcomes appear may or may not be important:

  Case 1: Sampling with replacement, the order is important
  Case 2: Sampling without replacement, the order is important
  Case 3: Sampling without replacement, the order is not important
  Case 4: Sampling with replacement, the order is not important

- *r*-tuples denoted by $(z_1, z_2, \cdots, z_r)$ will be used to describe the outcomes.

Counting Methods

Case 1: Sampling with replacement, the order is important

- All the possible outcomes are of the form $(z_1, z_2, \cdots, z_r)$.
- On the first draw, there are N possible objects; on the second draw there are N possible objects, and so on.
- The total number of possible outcomes is

$$\boxed{\underbrace{N \times N \times \cdots \times N}_{r \text{ times}} = N^r}$$

Counting Methods

Case 2: Sampling without replacement, the order is important

- All the possible outcomes are of the form $(z_1, z_2, \cdots, z_r)$, where all z's must be different.
- On the first draw, there are N; on the second there are N-1 possible objects, etc.
- The total number of possible outcomes is

$$(N)_r = N \times (N-1) \times \cdots \times (N-r+1).$$

- r ! is the number of ways that r objects can be arranged. These arrangements are called the *permutations*. Note that

$$\boxed{(N)_r = \frac{N!}{(N-r)!}}$$

Counting Methods

Case 3: Sampling without replacement, the order is not important

- All the possible outcomes are of the form $(z_1, z_2, \cdots, z_r)$, where all z's must be different.

- There are $r!$ orderings (permutations) of the chosen objects. We are interested in the number of distinct outcomes. Therefore, all permutations are considered as a single *combination*.

- To find the number of combinations, the total number of outcomes should be divided by the number of permutations.

$$\boxed{\binom{N}{r} = \frac{(N)_r}{r!} = \frac{N!}{(N-r)!\,r!}}$$

Counting Methods

Case 4: Sampling with replacement, the order is not important

- In order to show this case, we can form a table with one row whose columns separated by the symbol / correspond to an object. In a given draw, we place an x in the column corresponding to the object drawn.

- For example, consider the case of drawing 4 objects from 3 distinct objects with replacement. One possible outcome is

Object 1		Object 2		Object 3
xx	/		/	xx

- In other words, object 2 is not selected, objects 1 and 2 are selected twice. We can show this particular case with xx//xx.

Counting Methods

- In the general case case, the numbers of / and x will be *N*-1 and *r*, respectively.
- The problem is then is reduced to choosing *r* (if x is considered) or *N* - 1 (if / is considered) objects out of $N - 1 + r$ objects (the order is not important)
- We know that the result can be calculated by the combination:

$$\boxed{\binom{N-1+r}{r} = \binom{N-1+r}{N-1}}$$

Counting Methods

Example: An urn has r red balls and N - r black balls. If two balls are chosen with replacement, what is the probability of a red ball followed by a black ball?

Red balls: $\{1,2,\cdots,r\}$, Black balls: $\{r+1, r+2, \cdots, N\}$ (LABELING!)

The sample space: $S = \{(z_1, z_2):\ z_1, z_2 = 1,2,\cdots,N\}$

The given event: $E = \{(z_1, z_2):\ z_1 = 1,2,\ \cdots, r,\ z_2 = r+1, r+2, \cdots, N\}$

The total number of 2-tuples in S: $N_S = N \times N$

The total number of 2-tuples in E: $N_E = r \times (N-r)$

$$P[E] = \frac{N_E}{N_S} = \frac{r \times (N-r)}{N^2} = \frac{r}{N}\left(1 - \frac{r}{N}\right)$$

Counting Methods

Example: An urn has r red balls and N - r black balls. If two balls are chosen without replacement, what is the probability of a red ball followed by a black ball?

The sample space: $S = \{(z_1, z_2)\colon z_1 = 1,2,\cdots,N, z_2 \neq z_1\}$

The given event: $E = \{(z_1, z_2)\colon\ z_1 = 1,2,\ \cdots, r,\ z_2 = r+1, r+2, \cdots, N\}$

The total number of 2-tuples in S: $N_S = N \times (N-1)$

The total number of 2-tuples in E: $N_E = r \times (N-r)$

$$P[E] = \frac{N_E}{N_S} = \frac{r \times (N-r)}{N \times (N-1)} = \frac{r}{N}\frac{N-r}{N}\frac{N}{N-1}$$

Counting Methods

Example: Five balls numbered 1,2,3,4,5 are drawn from an urn without replacement. What is the probability that they will be drawn in the same order as their number?

$S = \{(z_1, z_2, z_3, z_4, z_5)\colon z_1 = 1,2,\cdots,5,\ z_2 \neq z_1,\ z_3 \neq z_2,\ z_4 \neq z_3,\ z_5 \neq z_4\}$

$E = \{(z_1, z_2, z_3, z_4, z_5)\colon\ z_1 = 1,\ z_2 = 2,\ z_3 = 3,\ z_4 = 4,\ z_5 = 5\}$

The total number of 5-tuples in *S*: $N_S = 5 \times 4 \times 3 \times 2 \times 1 = 120$

The total number of 2-tuples in *E*: $N_E = 1$

$$P[E] = \frac{N_E}{N_S} = \frac{1}{120}$$

Counting Methods

Example: N students are enrolled in a probability class. What is the probability that at least two students will have the same birthday?

Consider an urn that contains 365 balls labeled with a different day of the year. A ball is selected, its date is noted and it is returned to the urn. The probability desired is of the event that two or more balls selected are the same.

$E \stackrel{\text{def}}{=}$ {at least 2 students have same birthday}
$E^c =$ {no students have same birthday}

$$S = \{(z_1, \cdots, z_N): z_i = 1,2, \cdots, 365, i = 1,2, \cdots, N\}$$

$$E^c = \{(z_1, \cdots z_N): \ z_1 = 1, \cdots, 365, \ z_2 \neq z_1, \ z_3 \neq z_2, \cdots, \ z_N \neq z_{N-1}\}$$

$$P[E] = 1 - P[E^c] = 1 - \frac{N_{E^c}}{N_S} = 1 - \frac{365 \times 364 \times \cdots \times (365 - N + 1)}{365^N}$$

Binomial Probability Law

- The outcomes of an experiment are *success* and *failure* with probabilities p and 1-p, respectively. What is the probability of k successes out of a maxiumum of M successes?

- Consider an urn containing N balls of which N_R are red and N_B are black.

- If we associate the drawing of a red ball as a *success* and the drawing of a black ball as a *failure*, the required probability is equal to the probability of k red balls in M drawings without replacement.

Red balls: $\{1,2,\cdots,N_R\}$, Black balls: $\{N_R+1, N_R+2,\cdots,N\}$ (LABELING!)

The sample space: $S=\{(z_1,\cdots,z_M)$: $z_i=1,2,\cdots,M$ and no two z_i's are same$\}$

The event of interest:

$$E=\{(z_1,\cdots,z_M): k \text{ distinct intgers in } [1,N_R],\ M-k \text{ distinct integers in } [N_{R+1},N]\}$$

Binomial Probability Law

- The probability of obtaining *k* red balls is

$$P[k] = \frac{N_E}{N_S}$$

$$N_S = (N)_M \qquad N_E = \binom{M}{k}(N_R)_k(N_B)_{M-k} = M!\binom{N_R}{k}\binom{N_B}{M-k}$$

$$P[k] = \frac{M!\binom{N_R}{k}\binom{N_B}{M-k}}{(N)_M} = \frac{\binom{N_R}{k}\binom{N_B}{M-k}}{\binom{N}{M}}$$

- This law is called the *Hypergeometric law* and gives the probability of *k* successes when sampling without replacement is used.

- If sampling with replacement is used, the *Binomial law* results

Binomial Probability Law

- Instead of repeating the entire derivation, we will assume that N is sufficiently large, $M \ll N_R$ and $M \ll N_B$. As a result,

$$(N)_M \approx N^M, \qquad (N_R)_k \approx {N_R}^k, \qquad (N_B)_{M-k} \approx {N_B}^{M-k}$$

$$P[k] \approx \binom{M}{k} \frac{{N_R}^k {N_B}^{M-k}}{N^M} = \binom{M}{k} \left(\frac{N_R}{N}\right)^k \left(\frac{N_B}{N}\right)^{M-k}$$

- Using $N_R / N = p$ and $N_B / N = (N - N_R)/N = 1 - p$ gives the Binomial law

$$\boxed{P[k] = \binom{M}{k} (p)^k (1-p)^{M-k}}$$

Real World Example: Quality Control

Example: A memory chip manufacturer produces batches of 1000 chips for shipment to computer companies. 100 chips selected at random from a batch are tested. If 95 or more chips pass the test, the batch is acceptable, otherwise it defective.

Let us assume that a chip passes the test with probability p. What is the probability that a given batch will be acceptable?

The batch will not be defective if there are 95 or more successes out of 100 drawing. The probability of this occuring is

$$P[k \geq 95] = \sum_{k=95}^{100} \binom{100}{k} (p)^k (1-p)^{100-k}$$